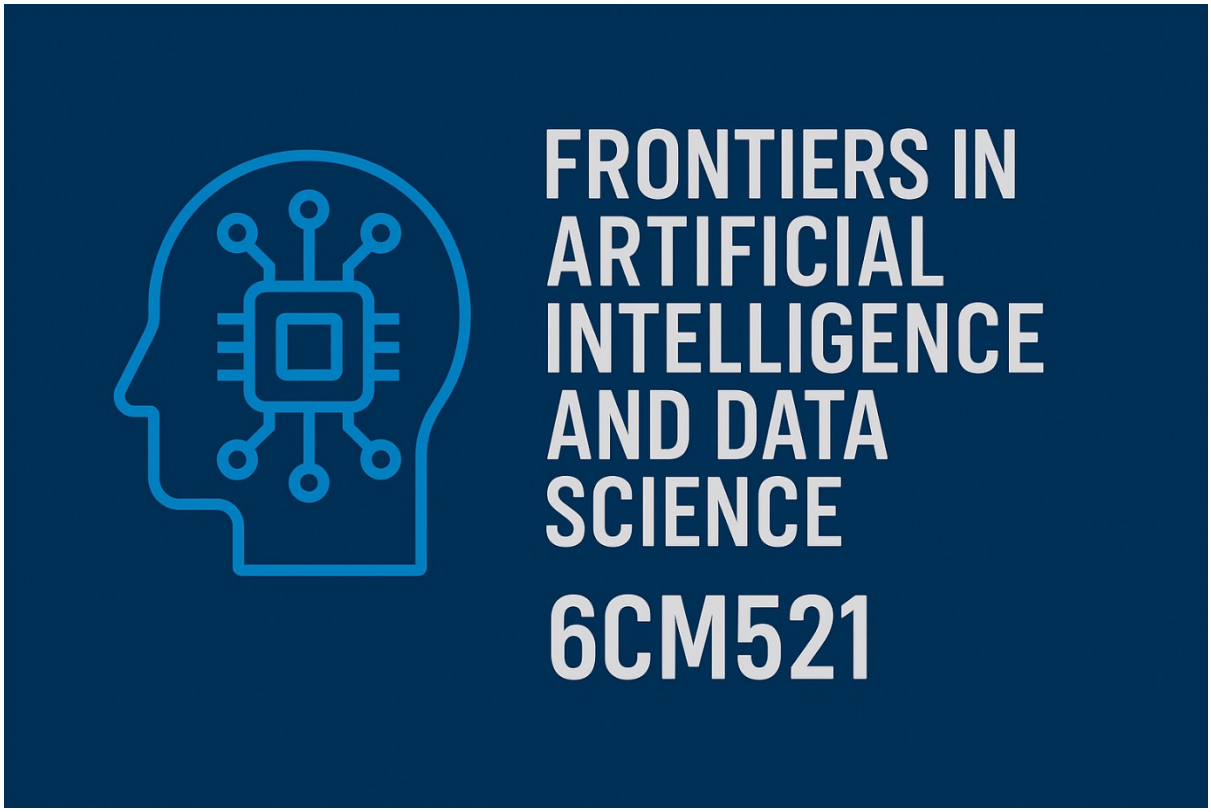
Frontiers in Artificial Intelligence and Data Science

6CM521



Assessment Brief

Associate Professor Harry Yu

Module Leader

- Harry Yu
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- Friday (10:00 – 11:00)

Key dates and details

Assessment Detail	Assessment Information
Assessment Mode:	Individual Artefact development and reporting/presentation
Assessment Weighting:	100%
Word count/Length:	Equivalent to 2000 words
Learning Outcomes:	1 and 2
Submission Method:	Blackboard Assignment
Submission Date:	4th May 2027
Provisional Feedback Release Date:	**18th May 2027**

Description of the assessment

Integrated AI Capstone Project (Two-Part Individual Assessment)

Overall Aim:
Enable each student to develop a specialised AI component—spanning vision, audio, language, or learning algorithm—that will later integrate into a powerful, multi-modal AI agent with image processing & generation, object recognition, sound recognition & synthesis, intelligent communication, and continual learning.

Weighting:

- Data Analysis, Implementation, Integration & Evaluation with presentation: 100% of final grade

Relationship to Programme Assessment Strategy

This module is designed for all programmes that related to Computing subjects and details are defined to suit the purpose.

Attributes and Skills

	Skills	Links to useful resources
☒	Critical thinking	Develop@Derby Skill Set
☒	Communication	Develop@Derby Skill Set
☒	Collaboration	Develop@Derby Skill Set
☒	Creative problem solving	Develop@Derby Skill Set
☒	Self-direction & planning	Develop@Derby Skill Set
☒	Numeracy, statistics & financial literacy	Develop@Derby Skill Set
☒	Digital	Develop@Derby Skill Set
☒	Resilience	Develop@Derby Skill Set
☒	Adaptability	Develop@Derby Skill Set
☒	Leadership & future thinking	Develop@Derby Skill Set

Assessment Content

A. Topic Options (choose 1 topic and datasets are provided):

1. Machine Vision for Deep Vein Thrombosis (DVT) Detection Using Ultrasound Videos

Venous thromboembolism, including deep vein thrombosis (DVT), is the third most common cause of vascular mortality worldwide, after heart attack and stroke. Prompt diagnosis of DVT is essential to reduce the risk of fatal complications.

Machine learning models have emerged as valuable tools for supporting timely diagnosis. By performing pixel-level segmentation of ultrasound videos, these models can assess vein compressibility, which is an important indicator of whether DVT is present.

However, creating pixel-level ground-truth annotations requires a considerable amount of manual effort. Therefore, this assessment aims to support the development of AI models for DVT detection using ultrasound videos without requiring exhaustive pixel-level annotations. The solution should reduce the burden of manual labelling while enabling robust and clinically relevant predictions.

2. Large Language Model-Based Dialogue System

This topic requires the development of a Large Language Model (LLM)-based dialogue system for a customer-support AI application. The aim is to create a system capable of understanding customer queries, generating appropriate responses and maintaining a coherent conversation flow.

B. Datasets

1. Machine Vision for Deep Vein Thrombosis (DVT) Detection Using Ultrasound Videos

Training: ThrombUS+ Ultrasound Dataset #1 - Compression Ultrasound Videos - Training Set | Zenodo

Testing: ThrombUS+ Ultrasound Dataset #1 - Compression Ultrasound Videos - Testing Set | Zenodo

2. Large Language Model based Dialogue System

bitext/Bitext-customer-support-llm-chatbot-training-dataset · Datasets at Hugging Face

C. Tasks: Data Analysis, Implementation, Integration & Evaluation tasks

Objectives

1. Demonstrate a thorough understanding of the dataset by conducting exploratory data analysis (EDA) for the selected AI topic, using appropriate statistical and visualisation methods.
2. Implement an AI algorithm or method that demonstrates the knowledge and skills required to apply cutting-edge technologies to develop a suitable solution.
3. Produce strong evaluation results that demonstrate critical thinking and provide evidence of the effectiveness and correctness of the developed solution.
4. Communicate and demonstrate the work effectively during a live presentation session.

Deadline: Week 21

D. **Deliverables:**

1. **Dataset Analysis — 30%**
   Complete the dataset analysis using Python, with appropriate Markdown comments and explanations included in the coding environment.
2. **Algorithm/Method Implementation and Evaluation — 40%**
   This should include:

- Research evidence supporting the solution design;
- Working code for the AI component, with clear instructions for setup and execution; and
- Baseline performance metrics evaluated using the test data.

3. **Embedded Reporting and Critical Analysis — 20%**
   Use Markdown comments within the coding environment to explain and critically analyse your work. This will form the reporting content reviewed by the examiner. A separate written report is not required.
4. **Final Demonstration and Presentation — 10%**

- Deliver a live demonstration and discussion lasting no more than 10 minutes.
- Presentation slides are not required.

The Python code for both the dataset analysis and algorithm implementation must be uploaded to the designated Blackboard submission point for marking. If no submission is made, no mark can be awarded. **Please do not upload the dataset you used to avoid submission failure due to large volume of data.**

Band	1. Dataset Analysis (30%)	2. Algorithm/Method Implementation and Evaluation (40%)	3. Technical Communication and Critical Analysis (20%)	4. Live Demo and Discussion (10%)
90–100	Exceptional and comprehensive EDA using 6+ purposeful statistical and visual analyses. Dataset structure, labels, quality, imbalance, bias, anomalies and topic-specific characteristics are critically examined. Preprocessing and data splitting are fully reproducible, with explicit prevention of leakage. Every important design decision is supported by EDA evidence.	Near state-of-the-art or exceptionally effective solution against suitable baselines. Fully reproducible implementation with clear dependencies, seeds and execution instructions. Reports 4+ appropriate evaluation measures, controlled comparisons and strong test performance. Includes detailed error analysis and ablation or component analysis.	Publication-quality technical narrative embedded throughout the notebook. Design decisions and results are supported by strong research evidence. Critical analysis thoroughly addresses limitations, ethics, safety, reliability and generalisability. Recommendations are specific and evidence-based; referencing is immaculate.	Exceptional, well-paced demonstration completed within 10 minutes. Clearly demonstrates the working system, inputs, outputs, live or saved metrics, and successful and unsuccessful cases. Questions are answered precisely, showing complete ownership and expert understanding.
80–89	Excellent EDA using 5+ relevant analyses. Data quality, distributions, imbalance and preprocessing are investigated accurately. Training, validation and testing arrangements are reproducible and leakage is appropriately controlled. Strong links are established between findings and solution design.	Excellent, reliable solution demonstrating meaningful improvement over a sensible baseline. Reproducible code with clear setup and strong testing. Reports 3+ appropriate evaluation measures, comparative results and several failure cases. Only minor technical or experimental limitations remain.	Excellent technical narrative with clear structure, strong research support and insightful interpretation. Limitations, errors, ethics and possible improvements are critically discussed. Referencing is accurate and consistent.	Excellent demonstration of a working system and key results. Delivery is clear, confident and within the time limit. Successful and unsuccessful examples are discussed. Questions demonstrate very strong technical understanding.
70–79	Strong EDA using 4+ relevant analyses. Most important data characteristics, quality issues and class or category distributions are correctly investigated. Preprocessing and splitting are justified. Findings clearly influence several solution-design decisions.	Strong working implementation performing well against an appropriate baseline. Reports 3+ relevant evaluation measures and includes meaningful test results and error analysis. Code is reproducible with only minor gaps in documentation, testing or experimental control.	Very good and coherent technical narrative. Most design decisions and results are explained and supported by appropriate references. Limitations and ethical issues are acknowledged and realistic improvements are proposed.	Very good demonstration covering the problem, method, main functionality and evaluation results. Most questions are answered accurately. Minor technical or timing issues do not affect the overall demonstration.
60–69	Good EDA using 3+ relevant analyses. Main features, labels, distributions and some data-quality issues are examined. Preprocessing and data splitting are mostly appropriate, although justification or connections to model design lack depth.	Good working solution with clear execution instructions. Includes a baseline or meaningful comparison and reports at least 2 relevant evaluation measures. Some failure cases are examined, but testing, comparison or critical evaluation is limited.	Good technical narrative explaining the approach, implementation and main results. Includes relevant references, limitations and some critical reflection, although parts remain descriptive or insufficiently linked to evidence.	Good demonstration showing the main functionality and results. Explanation is generally clear, but technical depth, failure analysis or responses to questions are limited.

50–59	Competent but mainly descriptive EDA using at least 2 meaningful analyses. Basic data quality, distributions and preprocessing are covered. Imbalance, bias, leakage or representativeness receives limited attention. Links between EDA and model design are weak.	Functional solution meeting the main task requirements. Baseline metrics and at least 2 evaluation measures are reported. Setup is mostly clear, but improvement over the baseline, reproducibility, testing or error analysis is limited.	Satisfactory technical narrative covering the main development stages. Some results and limitations are discussed, but critical comparison, research evidence, ethical consideration or referencing is limited.	Satisfactory demonstration showing that the main solution works. Core results are presented, but the explanation, evidence, organisation or answers to questions lack detail.
40–49	Basic EDA containing at least 1 meaningful statistic and 1 relevant visualisation. Some valid preprocessing is present, but the analysis is narrow, weakly interpreted or partly incorrect. Data splitting and leakage prevention are insufficiently justified.	Basic or partially effective implementation producing some credible outputs. Uses at least 1 valid evaluation measure, but the baseline, testing process or interpretation is weak. The solution relies heavily on standard code with limited adaptation.	Basic Markdown commentary describes the development process and outputs. The work lacks structure, critical interpretation or sufficient research evidence. Discussion focuses primarily on what was done rather than why it was appropriate.	Basic demonstration of partial functionality or credible saved outputs. Some aspects are explained correctly, but the student has difficulty justifying important decisions or answering technical questions.
30–39	Limited and poorly structured data investigation. Statistics or visualisations are minimal, inappropriate or largely unexplained. Important data-quality, preprocessing and splitting issues are overlooked.	Incomplete or unreliable implementation with major defects. Parts of the pipeline work, but the solution does not consistently meet the task requirements. Evaluation is minimal, inappropriate or based mainly on training results.	Limited and poorly organised commentary. Important methods, decisions and results are not adequately explained. Conclusions are generic or unsupported, with weak referencing and little critical reflection.	Ineffective or incomplete demonstration. Functionality is unreliable, results are unclear, and responses reveal significant gaps in technical understanding.
20–29	Very limited evidence of dataset understanding. Analysis mainly displays raw samples or automatic summaries without meaningful interpretation. Serious preprocessing, splitting or data-leakage problems are present.	Fragmented implementation containing substantial technical errors. Little credible evidence of functionality is provided. There is no valid baseline and the reported evaluation does not support the conclusions.	Very little meaningful reporting. Markdown mainly contains headings, short code descriptions or unsupported statements. Results, limitations and ethical issues are not meaningfully examined.	Minimal demonstration with little evidence that the student can operate or explain the submitted solution. Most technical questions cannot be answered satisfactorily.
10–19	Little relevant analysis. Dataset structure, labels or prediction target are substantially misunderstood. Analysis is mostly incomplete, incorrect or unrelated to the selected problem.	Very little relevant implementation. Code is predominantly non-functional, copied without meaningful adaptation or unrelated to the stated solution. No credible evaluation is provided.	Technical narrative is extremely limited or largely irrelevant. There is insufficient evidence of understanding, interpretation or critical analysis. Sources and third-party materials are inadequately acknowledged.	Very limited demonstration. The submitted solution cannot be meaningfully demonstrated, and the student shows little understanding of the code, method or results.
0–9	No meaningful dataset analysis or no assessable evidence submitted.	No working solution, no relevant implementation or no assessable evaluation evidence.	No meaningful technical narrative or critical analysis.	No meaningful demonstrat

Assessment AI-assistance

In this assessment AI-assistance is permitted in the following ways:

Searching related learning resources to support the development of the solution

Review on useful technologies and their implementations.

In this assessment AI-assistance should not be used for:

Generating any contents including codes, presentations and documents

Generation of solution design and datasets

It is YOUR RESPONSIBILITY to check ALL information generated by generative AI tools. Any misuse of generative AI tools could be considered ethical academic misconduct (as per Academic Regulations, Academic Misconduct, Section J2). If in doubt, please consult your module leader.

Anonymous Marking

Submissions in Blackboard

You must submit your work using your **student number** to identify yourself, not your name. You must not use your name in the text of the work at any point. When you submit your work in Turnitin you must submit your student number within the assignment document and in the *Submission title* field in Turnitin. Guidance is available showing how to do this.

Assessment Regulations

The University's regulations, policies and procedures for students define the framework within which teaching and assessment are conducted. Please make sure you are familiar with these regulations, policies and procedures.